PHYS3071

Astrophysics

Course Notes

Author: *automatoc-ti*

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Introduction

About These Notes

These notes accompany the PHYS3071 Astrophysics course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in special relativity, basic linear algebra, and basic calculus.

The material covers different cosmological topics, including but not limited to the Friedmann equations, and the energy budget of the ΛCDM model. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.

If you find any errors or have suggestions for improvement, please feel free to contact me at `yhip@connect.ust.hk`. You can check the latest version of the notes on GitHub: `https://github.com/AuTomatoc-Ti/PHYS3071-astro`.

Document Conventions

Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

Physics Intuition

Purpose: Builds conceptual understanding through plain language explanations.
This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

Theory

Purpose: Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

Question Examples

Purpose: Demonstrates practical application through worked problems.
Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

Foundation

Purpose: Supplies background knowledge and detailed proof steps that underpin the main content.
This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

Use this when: A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

Extra

Purpose: Provides supplementary material that extends *beyond* the course syllabus.
This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

Use this when: You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

IMPORTANT NOTE

Purpose: Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.
An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

Treat these as mandatory reading regardless of which reading path you follow.

REMARK

Purpose: Highlights a useful observation, connection, or naming convention.
A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

Use this when: Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

Mathematical Conventions

Key symbols.

Symbol	Meaning	Typical units
$M_\odot$	Solar mass	kg
$R_\odot$	Solar radius	m
$L_\odot$	Solar luminosity	W
T_{eff}	Effective temperature	K
μ	Mean molecular weight	—
ρ	Mass density	$\mathrm{kg\,m^{-3}}$

Remark

Unless otherwise stated, stellar quantities are quoted in solar-normalised form (for example $M/M_\odot$ and $L/L_\odot$), which makes scaling relations easier to compare across different stellar masses.

differential notation. We use $\dot{x} \equiv \frac{dx}{dt}$ for time derivatives of some variables $x(t)$.

CHAPTER 1

Basic Concepts in Astrophysics

1.1 Hydrostatic Equilibrium

1.2 Virial Theorem and Jeans Criterion

1.2.1 Energetics of Self-Gravitating Gas

THEORY

$$E_{\mathrm{GR}} = -\int_0^M \frac{Gm}{r}\,dm = -f_{\mathrm{GR}}\frac{GM^2}{R}, \tag{1.1}$$

$$\langle P\rangle = -\frac{1}{3}\frac{E_{\mathrm{GR}}}{V}, \tag{1.2}$$

$$P = \frac{1}{3}nm\langle v^2\rangle = \frac{2}{3}n\langle E_k\rangle. \tag{1.3}$$

For a star with a uniform density profile (i.e., ρ = constant), $f_{\mathrm{GR}} \approx 3/5$.

1.2.2 Jeans Instability

THEORY

$$|E_{\mathrm{GR}}| > 2E_{\mathrm{th}} \quad \text{(collapse)}, \tag{1.4}$$

$$M_J = \left(\frac{5k_BT}{G\mu m_H}\right)^{3/2}\left(\frac{3}{4\pi\rho}\right)^{1/2}, \tag{1.5}$$

$$\lambda_J = \sqrt{\frac{\pi c_s^2}{G\rho}}, \tag{1.6}$$

$$M_J \propto T^{3/2}\rho^{-1/2}. \tag{1.7}$$

IMPORTANT NOTE

Lower temperature and higher density both reduce M_J. This is why cooling strongly promotes fragmentation and cluster formation.

CHAPTER 2

Matter and Radiation

CHAPTER 3

Heat Transfers in Stars

Three main mechanisms of heat transfer in stars:

1. Conduction
2. Convection
3. Radiation (by photons and neutrinos from core)

CHAPTER 4

Thermonuclear Fusion in Stars

Recall that there are three primary energy sources in stars: gravitational contraction, radioactive decay and nuclear fusion.

In this chapter, we will focus on the process of thermonuclear fusion, which is the viable long-term stellar energy source in most stars.

4.1 Nuclear Basics

4.1.1 Nuclear Size and Binding Energy

THEORY

$$R = R_0 A^{1/3}, \qquad R_0 \approx 1.2 \times 10^{-15}\,\mathrm{m}, \tag{4.1}$$

$$BE(Z,N) = \left[Z m_p + N m_n - M(Z,N)\right] c^2. \tag{4.2}$$

Binding energy per nucleon increases for light nuclei and decreases for very heavy nuclei due to Coulomb repulsion.

4.1.2 Why Fusion Powers Stars

PHYSICS INTUITION

Stars gain energy when fusing light nuclei toward higher binding energy per nucleon. This trend reverses near iron, where further fusion is endothermic.

4.2 Hydrogen and Helium Burning

4.2.1 Proton-Proton (pp) Chain

FOUNDATION

Representative net reaction:

$$4\,{}^1\mathrm{H} \rightarrow {}^4\mathrm{He} + 2e^+ + 2\nu_e + \text{energy}. \tag{4.3}$$

Total released energy is about 26.7 MeV, with part carried away by neutrinos.

4.3 Barrier Penetration and Reaction Rates

4.3.1 Coulomb Barrier and Tunneling

Derivation

FOUNDATION

The **Gamov energy** is the minimum kinetic energy that two nuclei must have in order to overcome the Coulomb barrier and undergo fusion. It is given by the formula:

$$E_G = (\pi \alpha Z_1 Z_2)^2 2 m_r c^2 \tag{4.4}$$

where m_r is the reduced mass of the two nuclei, α is the fine-structure constant, and Z_1 and Z_2 are the atomic numbers of the two nuclei.

FOUNDATION

Classically, nuclei with $E < V_C$ cannot fuse. Quantum tunneling gives non-zero penetration probability P for $E < V_C$. The probability of tunneling through the Coulomb barrier can be approximated by the formula:

$$P \sim \exp\left[-\sqrt{\frac{E_G}{E}}\right], \tag{4.5}$$

where E_G is the Gamow energy.

4.4 Thermonuclear Reaction Rates

THEORY

$$\langle \sigma v \rangle = \left(\frac{8}{\pi \mu}\right)^{1/2} \frac{1}{(k_B T)^{3/2}} \int_0^\infty S(E) \exp\left[-\frac{E}{k_B T} - \sqrt{\frac{E_G}{E}}\right] dE. \tag{4.6}$$

The integrand peaks at the Gamow energy window, balancing high-energy tail abundance and tunneling probability.

THEORY

$$R_{AB} \propto T^a \tag{4.7}$$

where $a = \sqrt[3]{\frac{E_G}{4kT}}$ is a positive exponent that depends on the specific reaction and the temperature range.

CHAPTER 5

Stellar Structure

CHAPTER 6

Practice Questions & Solutions

6.1 Chapter 1: Basic Concepts in Astrophysics

6.1.1 Questions

1.1 Estimate the hydrodynamical free fall timescale for the Sun, a red giant of $M = M_\odot$ and $R = 100R_\odot$, and a white dwarf of $M = M_\odot$ and $R = 0.01R_\odot$ and $R = 100R_\odot$, and a white dwarf of $M = M_\odot$ and $R = 0.02R_\odot$.

1.2 Suppose that a star of mass M and radius R has a density distribution $\rho(r) = \rho_c(1 - \frac{r}{R})$, where ρ_c is the density at the center of the star.

(a) Find an expression for ρ_c in terms of M and R.

(b) Calculate the mass $m(r)$ interior to radius r.

(c) Calculate the total gravitational binding energy of the star.

(d) Find an expression for the pressure $P(r)$ as a function of r, assuming that the star is in hydrostatic equilibrium and that the pressure at the surface $(r = R)$ is zero.

(e) Assume that the material in the star is a **monatomic** ideal gas. Calculate the total internal energy of the star from $P(r)$, and show that the virial theorem is satisfied.

1.3

6.2 Chapter 2: Matter and Radiation

6.3 Chapter 3: Heat Transfers in Stars

6.4 Chapter 4: Thermonuclear Fusion in Stars

6.4.1 Questions

4.1 In the simplest model of α decay, the α particle is pre-formed and trapped within the nucleus by a potential similar to a Coulomb barrier. The mean decay rate λ is the frequency ν (with which the α particle hits the barrier) times the probability of penetration of the barrier (quantum tunneling probability).

(a) Write the approximate expression for the decay rate in terms of ν, E_G, and E (where E is the energy released by α decay).

(b) Find the Gamow energy of the α decay of ^{235}U.

(c) Find the Gamow energy of the α decay of ^{239}Pu.

(d) Assuming that the frequency ν is approximately the same for both nuclei, find the ratio

$$\frac{\lambda(^{239}\mathrm{Pu})}{\lambda(^{235}\mathrm{U})}$$

using $E(^{235}\mathrm{U}) = 4.68\mathrm{MeV}$ and $E(^{239}\mathrm{Pu}) = 5.24\mathrm{MeV}$. The lifetime of a radioactive species is inversely proportional to its decay rate. Given that the observed lifetime of ^{235}U is $\tau(^{235}\mathrm{U}) = 7.1 \times 10^8 y$, estimate the lifetime of ^{239}Pu. Compare your result with the experimental value

$$\tau(^{239}\mathrm{Pu}) = 2.41 \times 10^4 y$$

and comment on the accuracy of this simple model.

4.2 The **triple alpha process** is a set of nuclear reactions that occur in stars to produce carbon and other elements. Which of the following correctly describes the three steps of the triple alpha process, along with whether energy is absorbed or released during each step?

(a) $^4\mathrm{He} +^4 \mathrm{He} \rightarrow^8 \mathrm{Be}$

(b) $^8\mathrm{Be} +^4 \mathrm{He} \rightarrow^{12} \mathrm{C}^*$

(c) $^{12}\mathrm{C}^* \rightarrow^{12} \mathrm{C} + \gamma$

CHAPTER A

Declaration of AI Assistance

A.1 Use of Artificial Intelligence in This Document

This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

Author Verification: All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

CHAPTER B

Mathematical Tools & Derivations

B.1 Useful Identities

Trigonometric

$$\sin^2\theta + \cos^2\theta = 1, \quad \tan\theta = \sin\theta/\cos\theta, \quad \sin(A \pm B) = \sin A\cos B \pm \cos A \sin B.$$

Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Common Integrals

$$\int x^n\,dx = \tfrac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int e^{ax}\,dx = \tfrac{1}{a}e^{ax} + C,$$
$$\int \frac{dx}{x} = \ln|x| + C, \qquad \int \sin x\,dx = -\cos x + C.$$

Gamma–Zeta Integrals

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1}\,dx = \Gamma(s)\zeta(s), \qquad \Re(s) > 1,$$
$$\int_0^\infty \frac{x^{s-1}}{e^x + 1}\,dx = (1 - 2^{1-s})\Gamma(s)\zeta(s), \qquad \Re(s) > 0.$$

$$\int_0^\infty \frac{x^2}{e^x - 1}\,dx = 2\zeta(3), \qquad \int_0^\infty \frac{x^2}{e^x + 1}\,dx = \tfrac{3}{2}\zeta(3),$$
$$\int_0^\infty \frac{x^3}{e^x - 1}\,dx = \frac{\pi^4}{15} = 6\zeta(4), \qquad \int_0^\infty \frac{x^3}{e^x + 1}\,dx = \frac{7\pi^4}{120} = \tfrac{7}{8}\,6\zeta(4).$$

Taylor Expansions

$$f(x) = f(a) + f'(a)(x-a) + \tfrac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$\begin{aligned}
\sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \tfrac{x^3}{3!} + \tfrac{x^5}{5!} - \cdots \\
\cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \cdots \\
e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \tfrac{x^2}{2!} + \cdots \\
\sqrt{1+x} &= \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \cdots
\end{aligned}$$

Common Approximations

Small-argument limits ($|x| \ll 1$):

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}, \qquad \tan x \approx x,$$
$$\sinh x \approx x, \qquad \cosh x \approx 1 + \frac{x^2}{2}, \qquad \tanh x \approx x.$$

Large positive argument ($x \gg 1$):

$$\sinh x \approx \frac{1}{2}e^x, \qquad \cosh x \approx \frac{1}{2}e^x, \qquad \tanh x \approx 1.$$

Large negative argument ($x \ll -1$):

$$\sinh x \approx -\frac{1}{2}e^{-x}, \qquad \cosh x \approx \frac{1}{2}e^{-x}, \qquad \tanh x \approx -1.$$

B.2 Relativistic Limits

The energy-momentum relation is $E^2 = p^2c^2 + m^2c^4$. So in a relativistic limit, we can have:

$$\{E, m, \rho\} \ll \{p, T\}$$

which any one of the values in left-hand set is much smaller than any one of the values in the right-hand set.

CHAPTER C

Constants & Conversion Factors

C.1 Physical Constants

Quantity	Symbol	Value (SI)
Speed of light	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{m^{3}\,kg^{-1}\,s^{-2}}$
Hubble constant today	H_0	$67.8 \pm 0.77\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Cosmological constant	Λ	$1.11 \times 10^{-52}\,\mathrm{m^{-2}}$
Cosmological constant	Λ	$2.036 \times 10^{-35}\,\mathrm{s^{-2}}$
Planck constant	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s}$
Boltzmann constant	k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Fermi constant	G_F	$1.166 \times 10^{-5}\,\mathrm{GeV^{-2}}$
Fine-structure constant today	α	$1/137.036$
Planck mass	M_{Pl}	$2.176 \times 10^{-8}\,\mathrm{kg}$

C.2 Conversion Factors

Astronomical Units

Conversion	Value
1 astronomical unit (AU)	$1.496 \times 10^{11}\,\mathrm{m}$
1 light-year (ly)	$9.461 \times 10^{15}\,\mathrm{m}$
1 parsec (pc)	$3.086 \times 10^{16}\,\mathrm{m}$
1 parsec (pc)	$3.262\,\mathrm{ly}$
1 year (yr)	$3.156 \times 10^{7}\,\mathrm{s}$
1 Solar mass ($M_\odot$)	$1.989 \times 10^{30}\,\mathrm{kg}$
1 solar radius ($R_\odot$)	$6.957 \times 10^{8}\,\mathrm{m}$
1 Earth mass ($M_\oplus$)	$5.972 \times 10^{24}\,\mathrm{kg}$
1 Earth radius ($R_\oplus$)	$6.371 \times 10^{6}\,\mathrm{m}$
CMB temperature (T_{CMB})	$2.725\,\mathrm{K}$
CνB temperature ($T_{\mathrm{C\nu B}}$)	$1.95\,\mathrm{K}$

Energy Units

Conversion	Value
1 eV	1.602×10^{-19} J
1 electron mass (m_e)	$0.511\,\mathrm{MeV}/c^2$
1 electron mass (m_e)	9.109×10^{-31} kg
1 proton mass (m_p)	$938.3\,\mathrm{MeV}/c^2$
1 proton mass (m_p)	1.673×10^{-27} kg
1 neutron mass (m_n)	$939.6\,\mathrm{MeV}/c^2$
1 neutron mass (m_n)	1.675×10^{-27} kg

C.3 SI Unit Prefixes

Prefix	Name	Factor
10^{-12}	pico (p)	10^{-12}
10^{-9}	nano (n)	10^{-9}
10^{-6}	micro (μ)	10^{-6}
10^{-3}	milli (m)	10^{-3}
10^{3}	kilo (k)	10^{3}
10^{6}	mega (M)	10^{6}
10^{9}	giga (G)	10^{9}
10^{12}	tera (T)	10^{12}

C.4 Nuclei Binding Energies & Atomic Masses

Nucleus	Atomic Mass (u)	B(MeV)	B/A (MeV/u)
e^-	0.00054858	–	–
p	1.0072765	–	–
n	1.0086649	–	–
^{1}H	1.0078250	0.000000	0.000000
^{2}H	2.0141018	2.224515	1.112258
^{3}H	3.0160493	8.481773	2.827258
^{3}He	3.0160293	7.718036	2.572679
^{4}He	4.0026032	28.295803	7.073951
^{6}Li	6.0151223	31.994603	5.332434
^{7}Li	7.0160040	39.244654	5.606379
^{8}Be	8.0053051	56.499667	7.062458
^{9}Be	9.0121821	58.165096	6.462788
^{12}C	12.0000000	92.162051	7.680171
^{13}C	13.0033550	97.108223	7.469863
^{14}N	14.0030740	104.658962	7.475640
^{15}N	15.0001090	115.492214	7.699481
^{16}O	15.9949150	127.619412	7.976213
^{17}O	16.9991320	131.762631	7.750743
^{18}O	17.9991610	139.806972	7.767054
^{20}Ne	19.9924400	160.645559	8.032278
^{24}Mg	23.9850420	198.257479	8.260728
^{28}Si	27.9769270	236.537285	8.447760
^{32}S	31.9720710	271.781333	8.493167
^{56}Fe	55.9349420	492.255833	8.790283

Where **B** is the total binding energy of the nucleus, and **B/A** is the average binding energy per nucleon.

Note: $u = 1.661 \times 10^{-27}\,\text{kg} = 931.5\,\text{MeV}/c^2$.

REFERENCES

[1] *PHYS3071 Astrophysics Lecture Notes*. 2026.

[2] Anthony C. Phillips. *The Physics of Stars*. 1st. Manchester Physics Series. Chichester: John Wiley & Sons, 1994. ISBN: 978-0-471-94057-9.